

Zero-Drop 5G: Cryptography-Powered Backup Connectivity via NTN

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1. Objective

This proposal aims to develop a **resilient, cryptographically secure failover system for 5G networks**, specifically designed for **highly critical applications** such as **defense communications and disaster recovery**. The solution leverages **Non-Terrestrial Networks (NTNs)**, such as satellite constellations, combined with **5G network slicing** to guarantee **uninterrupted, secure connectivity** even during network failures. By incorporating **Trusted Execution Environments (TEEs)** and **Zero-Knowledge Proofs (ZKPs)**, the system ensures **privacy-preserving validation** and secure backup network routing, facilitating **Zero-Drop 5G** for the most demanding scenarios.

2. Summary of the Proposal

In critical and sensitive scenarios, such as defense and disaster recovery, ensuring **uninterrupted communication** is essential. Our solution leverages **5G network slicing, NTN connectivity**, and advanced cryptographic mechanisms to guarantee secure failover during network disruptions. The architecture integrates **TEEs** for **secure attestation** and **ZKPs** for **privacy-preserving validation** of failover actions. A decentralized **blockchain** ensures **real-time verification** of NTN backup links and states, ensuring **mission-critical communication** remains uninterrupted. This solution provides **Zero-Drop 5G**, making it suitable for environments where trust, performance, and security are paramount.

3. Detailed Project Proposal

Problem Statement

In scenarios like **military operations, emergency response, and national security systems**, losing connectivity due to **disasters or cyber-attacks** can have devastating consequences. While **5G** offers high throughput and low latency, it's still vulnerable to network disruptions. **NTNs** (satellite constellations, HAPs) offer a potential backup but pose challenges in **trust, security, and verifiability**.

Proposed Solution

Our solution combines **5G network slicing** with **NTN backup** connectivity to **ensure seamless** failover when the primary terrestrial network fails. Key technologies include:

- **TEEs** for secure handling and validation of failover actions
- **ZKPs** for privacy-preserving attestation of backup links
- **Blockchain** for auditability and continuous validation of NTN links
- **Dynamic trust scoring** to select the best NTN link based on mission-specific requirements, routing NTN paths based on security and performance.

Technology Stack

- **5G Network Slicing**: Mission-specific slices for dedicated services (e.g., military, emergency).
- **TEEs**: Secure attestation of devices and failover actions.
- **ZKPs**: Privacy-preserving validation of NTN backup integrity.
- **Blockchain**: Decentralized network for continuous liveness validation and backup state auditing.

Prototype Plan

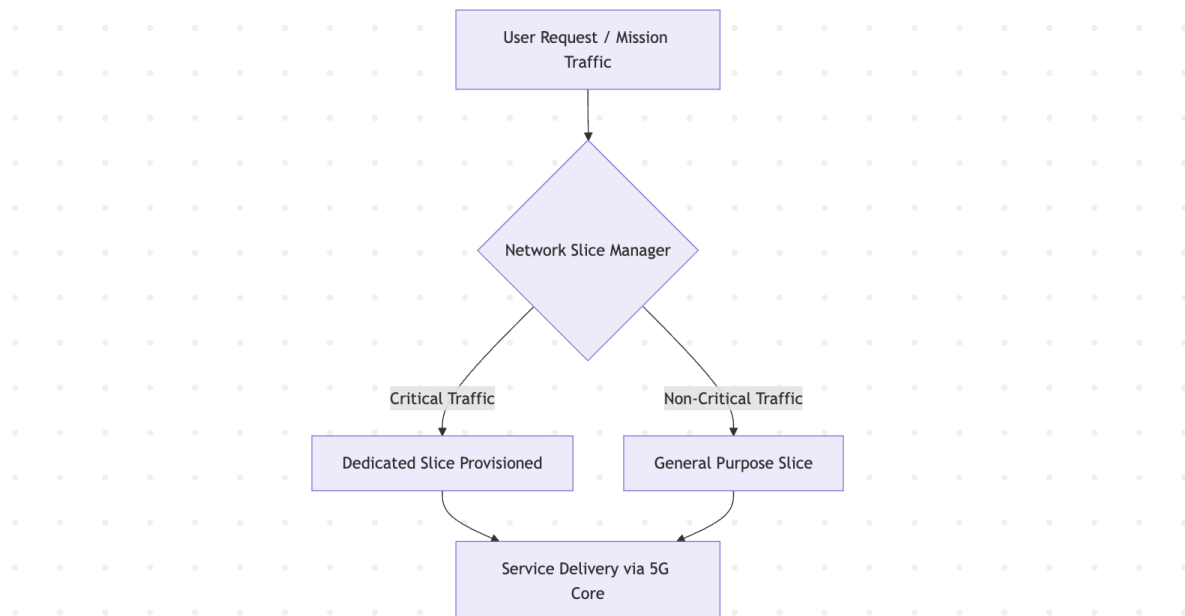


5G Network Slicing

[The 5G system](#) should support mobility between the supported access networks (e.g. NG-RAN, WLAN, fixed broadband access network, 5G satellite access network). A more advanced network slice can be elaborated with advancements in cryptographic primitives for modern user cases.

Using [Reticulum](#) (The cryptography-based networking stack for building unstoppable networks with LoRa, Packet Radio, WiFi and everything in between.) to design and implement an advanced 5G network slice with more cryptographic primitives capabilities.

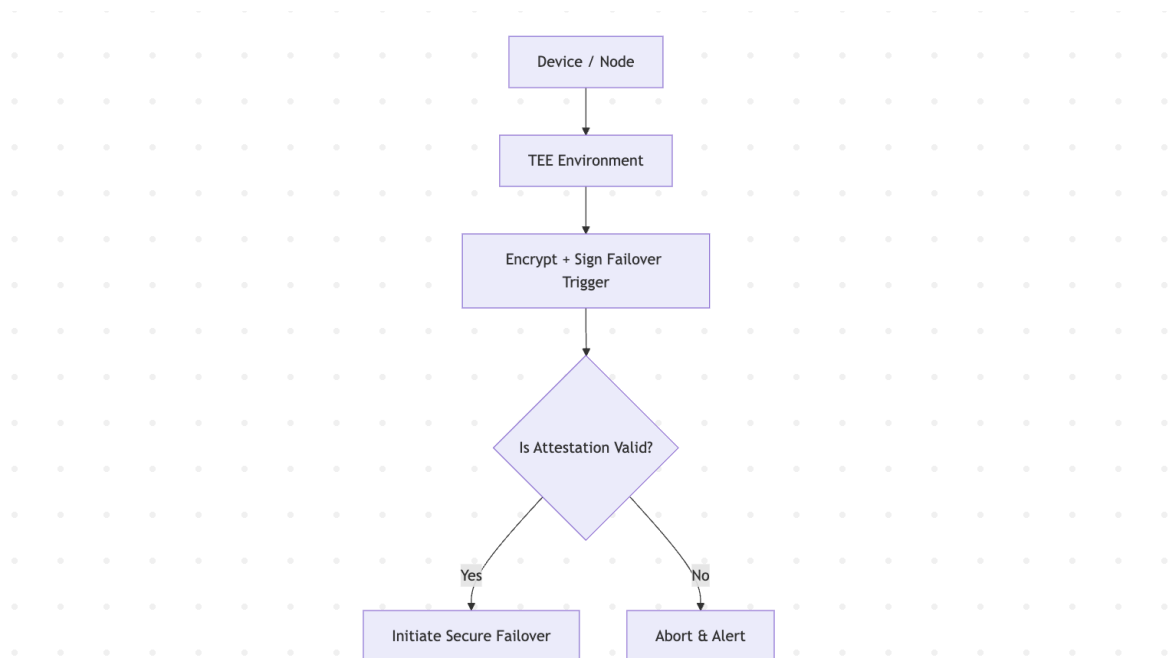
Based on the type of request (e.g., emergency), the slice manager assigns a dedicated or general-purpose 5G slice, optimizing performance and isolation.



TEE

TEEs monitor the network health and validate any failover actions which require backup NTN to be triggered. Only predetermined and validated/certified nodes serve as monitors and validators. Once a network failure is detected, a backup link is selected and a ZKP provides the proof of liveness and authenticity of the backup link. Once the NTN backup is validated, the node establishes an encrypted message and data exchange link on the NTN to facilitate critical communications.

NTN links, network attestations and message and data communications are simulated through [Cyptosim](#) which is a test environment implemented on the concepts of [SpaceTEE](#) by [Cryptosat](#).



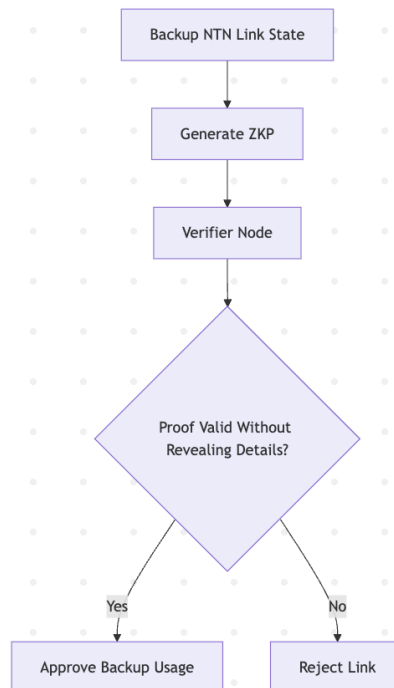
ZKP

Research in zero-knowledge proofs has been motivated by authentication systems where one party wants to prove its identity to a second party via some secret information (such as a password) but doesn't want the second party to learn anything about this secret. This is called a "zero-knowledge proof of knowledge". However, a password is typically too small or insufficiently random to be used in many schemes for zero-knowledge proofs of knowledge. A zero-knowledge password proof is a special kind of zero-knowledge proof of knowledge that addresses the limited size of passwords. More ZKPs application:

- Ethical behavior
- Blockchains
- Decentralised Identifiers and more

Here we can use [arkworks-rust based-groth16](#) (A rust based implementation of Groth16 zkSNARK),

ZKPs verify the integrity of backup NTN links without revealing sensitive state information.



Blockchain

A distributed database or ledger shared across a computer network's nodes. They are best known for their crucial role in cryptocurrency systems, maintaining a secure and decentralized record of transactions, but they are not limited to cryptocurrency uses. Blockchains can be used to make data in any industry immutable—meaning it cannot be altered.

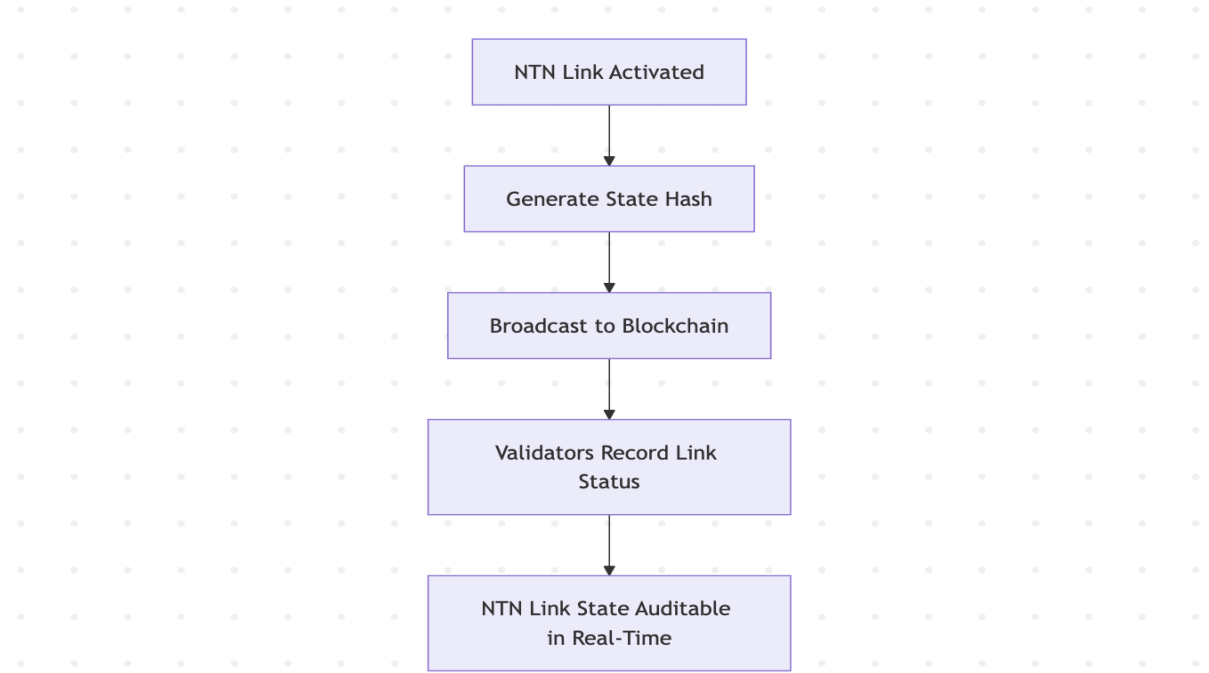
Since a block can't be changed, the only trust needed is at the point where a user or program enters data. This reduces the need for trusted third parties, such as auditors or other humans, who add costs and can make mistakes.

Here we can use [Ethereum Blockchain](#) as the core network layer to verify backups.

As shown the network effect of and decentralization of Ethereum below,



Blockchain provides transparency and publicly verifiable records. Every backup action is verifiably logged on a blockchain for auditability and transparency.



Team Composition

- Lead Cryptography Engineer (5G Network & NTN Integration, ZKPs)
- Blockchain Scientist (Decentralized Validator Network)
- Security Expert (TEEs, End-to-End Encryption & Secure Session Management)

4. Societal & Industrial Impact

Societal Impact

*The solution ensures **continuous communication** during **disasters** and **emergency recovery**, saving lives and supporting national security. It empowers **military** and **emergency services** with a robust, failproof communication system, ensuring **quick response** and **coordination** during crises.*

Industrial Impact

*By providing **uninterrupted connectivity**, this system can be adapted for various industries like **defense**, **energy**, and **critical infrastructure**. It also enhances the **reliability** of **5G networks**, fostering greater trust and investment in next-gen connectivity technologies.*

5. Scalability & Market Readiness

Scalability

*The solution is built to scale with **global NTN constellations** and **expanding 5G networks**. Its modular design allows it to be integrated into **different verticals** (e.g., defense, emergency services, industrial IoT) with varying service levels and requirements.*

Market Readiness

*With increasing demand for secure, reliable communication networks, the solution addresses the **growing need** for resilient networks in sectors like **defense**, **emergency response**, and **disaster recovery**. The system is ready for **pilot projects** in collaboration with government agencies and private industry partners.*

6. 5G Technology Utilization

5G Network Slicing

Enables **dedicated slices** for mission-critical services, ensuring optimal performance and security even during failover.

5G NTN Connectivity

Ensures **seamless failover** to **satellite constellations** or **high-altitude platforms (HAPs)**, providing **global backup connectivity** in case of terrestrial network failure.

Zero-Drop 5G

Continuous, uninterrupted communication with **high availability** and **low latency**, even under extreme conditions, made possible by **dynamic failover** via NTN.

Conclusion

The **Zero-Drop 5G** proposal presents a transformative approach to resilient connectivity by integrating **5G network slicing**, **Non-Terrestrial Networks**, and **advanced cryptographic technologies**. By ensuring seamless, secure failover during terrestrial network disruptions, the system addresses critical gaps in communication infrastructure for defense, disaster recovery, and other high-stakes applications. Through the use of **TEEs**, **ZKPs**, and **blockchain-based validation**, it establishes a trustworthy, scalable, and privacy-preserving backup connectivity layer. As the demand for dependable, always-on networks continues to rise, this solution stands at the forefront of next-generation communication—ensuring that when the stakes are highest, connectivity never drops.